How to run snl-bes-elm baseline firmware and software

How to install the Software With Anaconda

- Step 1: Go to https://www.anaconda.com/download to get the latest version of anaconda. Example steps for installing anaconda are included below:
  $ cd /opt/slac

  $ wget https://repo.anaconda.com/archive/Anaconda3-2024.06-1-Linux-x86_64.sh
  $ bash Anaconda3-2024.06-1-Linux-x86_64.sh

- Step 2: Use the following command to add anaconda to your environment. This can be added to your .bash_profile.

  $ source /opt/slac/anaconda3/etc/profile.d/conda.sh

  Note: Please source the conda environment path that was used during the setup of conda, as it might be different from the path shown above.

- Step 3: It is important to use the latest conda solver (Very long "install" setup, but will save you a lot of time later)

  $ conda config --set channel_priority strict
  $ conda install -n base conda-libmamba-solver
  $ conda config --set solver libmamba

- Step 4: Build the "snl_bes_elm" conda environment

  $ conda create -n snl_bes_elm -c conda-forge -c tidair-tag snl_bes_elm

How to update software conda env

Assumes you have completed "How to install the Software With Anaconda" section already

- Step 1: Load the snl_bes_elm conda environment

  $ source /opt/slac/anaconda3/etc/profile.d/conda.sh

  $ conda activate snl_bes_elm

- Step 2: Update the snl_bes_elm conda base on latest tag release

  $ conda update snl_bes_elm -c tidair-tag

How to reprogram the firmware on the KCU1500

Assumes you have completed "How to install the Software With Anaconda" section already

- Step 1: Load the snl_bes_elm conda environment

  $ source /opt/slac/anaconda3/etc/profile.d/conda.sh

  $ conda activate snl_bes_elm

- Step 2: Program the KCU1500 PROMs

  $ updatePcieFpga --path path_to_image_dir

- Step 3: reboot the machine

  $ sudo reboot

How to run the baseline python software

Assumes you have completed "How to install the Software With Anaconda" and "How to reprogram the firmware on the KCU1500" sections already

- Step 1: Load the snl_bes_elm conda environment

  $ source /opt/slac/anaconda3/etc/profile.d/conda.sh

  $ conda activate snl_bes_elm

- Step 2: Download the .pt data file

  $ wget <TBD.pt file>

- Step 3: Run the baseline python software

  $ SnlBesElm_LoadConfig --config <TBD.pt file>

Required Updates To Driver

- Edit files in the driver to match this diff:

  https://github.com/slaclab/aes-stream-drivers/compare/realtime...lock_cpu

  The important changes are in the **5** edits in **axis_gen2.c**

  The other changes are related to a kernel update which requires **pr_warning** to be replaced with **pr_warn**

  insmod datadev.ko cfgIrqDis=cpuNum cfgSize=1000000 cfgTxCount=1024 cfgRxCount=1024